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1 Gaussian channels

We consider a pure loss channel [1] $\Phi^{(\gamma)}$, characterized by matrices

$$X^{(\gamma)} = \cos\theta^{(\gamma)}\mathbb{1}, \quad Y^{(\gamma)} = \sin^2\theta^{(\gamma)}\mathbb{1}, \tag{1}$$

acting on first and second moments as follows:

$$\bar{\boldsymbol{r}} \mapsto X\bar{\boldsymbol{r}}, \tag{2}$$
$$\boldsymbol{\sigma} \mapsto X\boldsymbol{\sigma}X^{\mathrm{T}} + Y, \tag{3}$$

dual of which, $\Phi^{*(\gamma)}$, transforms the POVM of the ideal measurement into the POVM of the noisy measurements by changing respective covariance matrix $\Sigma_{\mathrm{id}}^{(\gamma)}$ into $\Sigma_m^{(\gamma)}$ as follows:

$$\Phi^{*(\gamma)}: \quad \Sigma_{\mathrm{id}}^{(\gamma)} \mapsto \Sigma_m^{(\gamma)} = \frac{1}{\cos^2\theta^{(\gamma)}}\Sigma_{\mathrm{id}}^{(\gamma)} + \tan^2\theta^{(\gamma)}\mathbb{1}, \tag{4}$$

where we used the expressions for the duals of X and Y:

$$X^* = X^{-1}, \quad Y^* = X^{-1}YX^{-1^{\mathrm{T}}} \tag{5}$$

For homodyne detection, covariance matrices of ideal and noisy measurements are given by [2]

$$\Sigma_{\mathrm{id}}^{(\mathrm{H})} = \lim_{r\to+\infty} R(\phi)\,\mathrm{diag}(e^{-2r}, e^{2r})\,R^{\mathrm{T}}(\phi) \tag{6}$$
$$\Sigma_m^{(\mathrm{H})} = \lim_{r\to+\infty} R(\phi)\,\mathrm{diag}(e^{-2r}+\sigma_N, e^{2r})\,R^{\mathrm{T}}(\phi) = \Sigma_{\mathrm{id}}^{(\mathrm{H})} + \Sigma_N^{(\mathrm{H})}, \tag{7}$$
$$\Sigma_N^{(\mathrm{H})} = R(\phi)\,\mathrm{diag}(\sigma_N, 0)\,R^{\mathrm{T}}(\phi). \tag{8}$$

As the expression are taking limits, implying that

$$\frac{1}{\cos^2\theta}\Sigma_{\mathrm{id}}^{(\mathrm{H})} \sim \Sigma_{\mathrm{id}}^{(\mathrm{H})}, \tag{9}$$
$$e^{2r} + \mathrm{const} \sim e^{2r}, \tag{10}$$

we may write the effective excess noise covariance matrix

$$\Sigma_N^{(\mathrm{H})\,\mathrm{eff}} = R(\phi)\,\mathrm{diag}(\sigma_N, \sigma_N)\,R^{\mathrm{T}}(\phi), \tag{11}$$

and solving for $\theta^{(\mathrm{H})}$ yields

$$\cos\theta^{(\mathrm{H})} = \sqrt{\frac{1}{\sigma_x}}, \quad \sin\theta^{(\mathrm{H})} = \sqrt{\frac{\sigma_N}{\sigma_x}}. \tag{12}$$

For double homodyne detection,

$$\Sigma_m^{(\mathrm{DH})} = R(\phi)\mathrm{diag}(\delta_1, \delta_2)R^{\mathrm{T}}(\phi) = \Sigma_{\mathrm{id}}^{(\mathrm{DH})} + \Sigma_N^{(\mathrm{DH})}, \tag{13}$$
$$\Sigma_{\mathrm{id}}^{(\mathrm{DH})} = R(\phi)\,\mathrm{diag}(e^{2r}, e^{-2r})\,R^{\mathrm{T}}(\phi) \tag{14}$$
$$\Sigma_N^{(\mathrm{DH})} = R(\phi)\,\mathrm{diag}(4\sigma_N^{(1)}(r), 4\sigma_N^{(2)}(r))\,R^{\mathrm{T}}(\phi), \tag{15}$$
$$4\sigma_N^{(1)}(r) = \delta_1 - e^{2r}, \tag{16}$$
$$4\sigma_N^{(2)}(r) = \delta_2 - e^{-2r}, \tag{17}$$
$$\delta_i = 2\sigma_i - 1. \tag{18}$$

First, we note that, since $\Sigma_m^{(\mathrm{DH})}$ is anisotropic, i.e. $\delta_1 \neq \delta_2$ generally, vacuum ancilla that defines Y (1) will not be suitable. Instead, consider arbitrary ancilla state Σ_{anc}, so that:

$$Y^{(\mathrm{DH})} = \sin^2\theta \Sigma_{\mathrm{anc}}. \tag{19}$$

Without loss of generality, we will consider the ancilla to be in pure state with $\det \Sigma_{\mathrm{anc}} = 1$,

$$\Sigma_{\mathrm{anc}} = R(\phi)\,\mathrm{diag}\left(x, \frac{1}{x}\right) R^{\mathrm{T}}(\phi), \quad x > 0, \tag{20}$$

so that the equation (4) takes the form

$$\Sigma_m^{(\mathrm{DH})} = \frac{1}{\cos^2\theta}\Sigma_{\mathrm{id}}^{(\gamma)} + \tan^2\theta \Sigma_{\mathrm{anc}}, \tag{21}$$

arriving at the system of equations

$$\begin{cases} \delta_1 = \frac{e^{2r}}{\cos^2\theta} + \tan^2\theta x \\ \delta_2 = \frac{e^{-2r}}{\cos^2\theta} + \tan^2\theta x^{-1} \end{cases}. \tag{22}$$

Solving the system for the elements of the ancilla covariance matrix, we arrive at quadratic equation

$$x^2 + bx + c = 0, \tag{23}$$

$$b = \frac{-\delta_1 e^{-2r} + \delta_2 e^{2r}}{4\sigma_N^{(2)}}, \quad c = \frac{-\sigma_N^{(1)}}{\sigma_N^{(2)}}, \tag{24}$$

$$4\sigma_N^{(1)} = \delta_1 - e^{2r}, \quad 4\sigma_N^{(2)} = \delta_2 - e^{-2r}, \tag{25}$$

with the solution given by

$$x(r) = \frac{1}{2}\left[-b + \sqrt{b^2 - 4c}\right]. \tag{26}$$

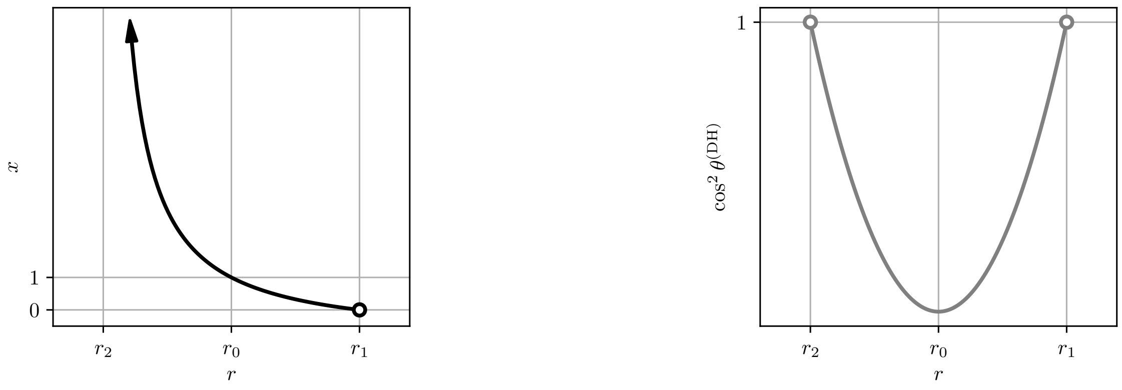
It's also worth noting that beyond the r interval solutions x become strictly unphysical, i.e. $x > 0 \iff r \in (r_2, r_1)$. This is due to $\sigma_N^{(i)}$ becoming negative, so that the discriminant becomes negative as well. At $r = r_1$ or $r = r_2$ ancilla is not well-defined, i.e. $x = 0$ or $x = \infty$, respectively.

Returning to $\cos^2\theta^{(\mathrm{DH})}$:

$$\cos^2\theta^{(\mathrm{DH})}(r) = \frac{e^{2r} + x(r)}{\delta_1 + x(r)}. \tag{27}$$

Dependence of x (26) and $\cos^2\theta^{(\mathrm{DH})}$ on squeezing parameter r is plotted on Fig. 1. While $x(r_0) = 1$ is a feature of $\delta_1 = \delta_2$, minimum of $\cos^2\theta^{(\mathrm{DH})}$ is always at $r = r_0$:

$$\cos^2\theta^{(\mathrm{DH})}(r_0) = \frac{s + \sqrt{\frac{s-\delta_1}{s^{-1}-\delta_2}}}{\delta_1 + \sqrt{\frac{s-\delta_1}{s^{-1}-\delta_2}}}, \quad s = \sqrt{\frac{\delta_1}{\delta_2}}. \tag{28}$$



(a) (b)

Figure 1: Dependence of functions defining attenuator channel that describes noisy double homodyne measurements on squeezing parameter r, see Eq. (26) and Eq. (27). $\delta_1 = \delta_2$.

2 GG02

We consider local action of dual to $\Phi^{*(\gamma)}$ channel, $\Phi^{(\gamma)}$, on second mode of TMSVS shared by Alice and Bob [3]:

$$\Sigma_{\mathrm{AB}}^{\mathrm{EB}} = \begin{pmatrix} V\mathbb{1} & c\sigma_z \\ c\sigma_z & V_{\mathrm{B}}\mathbb{1} \end{pmatrix}, \tag{29}$$

$$V = 1 + 4V_{\mathrm{A}}, \tag{30}$$

$$c = \sqrt{T_{\mathrm{ch}}(V^2 - 1)}, \tag{31}$$

$$V_{\mathrm{B}} = T_{\mathrm{ch}}(V - 1) + 1 + \xi_{\mathrm{ch}}, \tag{32}$$

where T_{ch} and ξ_{ch} are channel transmission and noise variance, respectively.

Using Eq. (1), we have

$$\Phi^{(\gamma)}(\Sigma_{\mathrm{AB}}^{\mathrm{EB}}) = \begin{pmatrix} V\mathbb{1} & c\cos\theta^{(\gamma)}\sigma_z \\ c\cos\theta^{(\gamma)}\sigma_z & \cos^2\theta^{(\gamma)}\left(V_{\mathrm{B}} + \tan^2\theta^{(\gamma)}\Sigma_{\mathrm{anc}}^{(\gamma)}\right) \end{pmatrix}, \tag{33}$$

mutual information is given by the expressions

$$I_{\mathrm{AB}}^{(\mathrm{H})} = \frac{1}{2}\log\left[1 + \frac{4TV_{\mathrm{A}}}{\sigma_x + \xi}\right], \tag{34}$$

$$I_{\mathrm{AB}}^{(\mathrm{DH})} = \frac{1}{2}\sum_i \log\left[1 + \frac{4TV_{\mathrm{A}}}{2\sigma_i + \xi}\right]. \tag{35}$$

For homodyne detection, from Eq. (12) we have

$$\Phi^{\mathrm{H}}\left(\Sigma_{\mathrm{AB}}^{\mathrm{EB}}\right) = \begin{pmatrix} V\mathbb{1} & \frac{c}{\sqrt{\sigma_x}}\sigma_z \\ \frac{c}{\sqrt{\sigma_x}}\sigma_z & \frac{1}{\sigma_x}\left(V_{\mathrm{B}} + \sigma_N\right)\mathbb{1} \end{pmatrix}. \tag{36}$$

Defining total channel transmission T that consists of channel transmission T_{ch} and detection losses $\frac{1}{\sigma_x}$, as well as scaled noise variance ξ,

$$T = T_{\mathrm{ch}} \cdot \frac{1}{\sigma_x}, \tag{37}$$

$$\xi \equiv \xi_{\mathrm{ch}} \cdot \frac{1}{\sigma_x}, \tag{38}$$

so that Bob's variance takes the form

$$\frac{V_{\mathrm{B}}+\sigma_N}{\sigma_x} = \frac{T_{\mathrm{ch}}(V-1)+\sigma_x+\xi_{\mathrm{ch}}}{\sigma_x} = T(V-1)+1+\xi \equiv \tilde{V}_{\mathrm{B}}, \tag{39}$$

and correlation remains valid,

$$\frac{c}{\sqrt{\sigma_x}} = \sqrt{\frac{T_{\mathrm{ch}}(V^2-1)}{\sigma_x}} = \sqrt{T(V^2-1)} \equiv \tilde{c}, \tag{40}$$

TMSVS covariance matrix (36) can be written as:

$$\Phi^{(\mathrm{H})}\left(\Sigma_{\mathrm{AB}}^{\mathrm{EB}}\right) = \begin{pmatrix} V\mathbb{1} & \tilde{c}\sigma_z \\ \tilde{c}\sigma_z & \tilde{V}_{\mathrm{B}}\mathbb{1} \end{pmatrix}. \tag{41}$$

It follows that for homodyne detection the effect of measurement asymmetry is equivalent to scaling both channel transmission (37) and channel noise (38) by factor σ_x^{-1}.

Symplectic eigenvalues of $\Phi_{\mathrm{H}}\left(\Sigma_{\mathrm{AB}}^{\mathrm{EB}}\right)$, $\nu_{1,2}^{(\mathrm{H})}$, can be computed using formulas [4]:

$$\nu_{1,2}^{(\mathrm{H})} = \frac{z \pm [\tilde{V}_{\mathrm{B}} - V]}{2}, \quad z = \sqrt{\left(V+\tilde{V}_{\mathrm{B}}\right)^2 - 4\tilde{c}^2}. \tag{42}$$

While channel noise is reduced in this setup, so is channel transmission, and, since transmission is multiplied by $V \gg 1$, as well as reduction in mutual information, this channel has negative effect, as seen from Fig. 2.

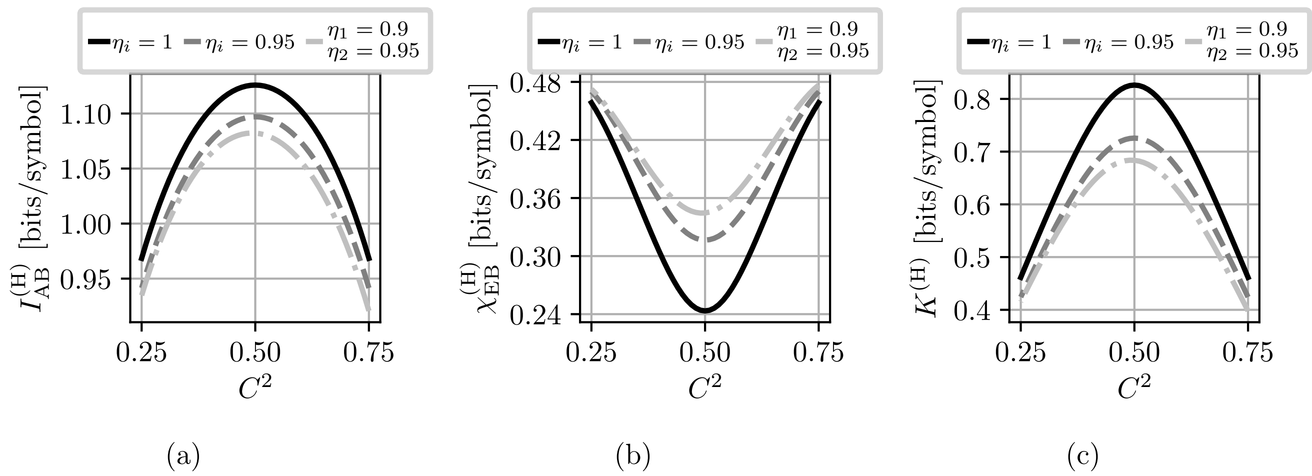


Figure 2: (a) Mutual information, $I_{\mathrm{AB}}^{(\mathrm{H})}$, (b) Holevo information, $\chi_{\mathrm{EB}}^{(\mathrm{H})}$ and (c) asymptotic secret fraction, $K^{(\mathrm{H})}$, as a function of the beam splitter transmission, C^2, of the homodyne receiver computed at different values of the detector efficiencies. The parameters are: $V=5$; $T_{\mathrm{ch}}=0.95$; $\xi_{\mathrm{ch}}=10^{-2}$; and $\beta=0.95$. As it is impossible to find r that maximizes $\chi_{\mathrm{EB}}^{(\mathrm{DH})}$ due to singularity, $r=r_0$.

From Fig. 3, it becomes apparent that the effect of imperfect measurements on maximal transmission distance is very slight. Even using exaggerated values of σ_x (correspond to $\eta=0.8$ and $\eta=0.5$) results in difference of about 1 km. However, bitrate is significantly reduced at low distances.

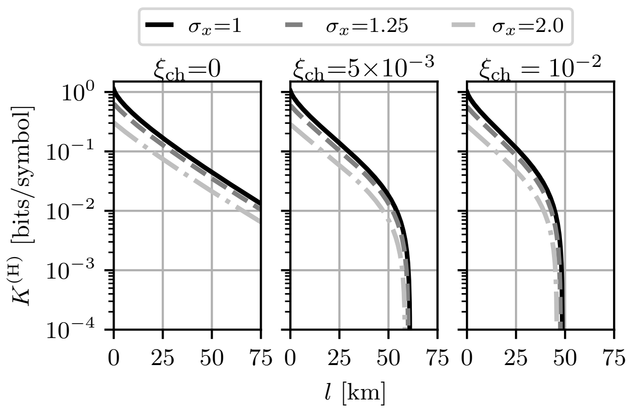


Figure 3: Asymptotic secret fraction $K^{(\mathrm{H})}$ versus channel length l, computed for various quadrature variances σ_x and channel noise variance ξ (as indicated in each panel title). The losses are assumed to be 20 dB per 100 km. The parameters are: $V = 5$ and $\beta = 0.95$.

For double homodyne detection, from Eq. (27) and Eq. (22), we have

$$\Phi^{(\mathrm{DH})}\left(\Sigma_{\mathrm{AB}}^{\mathrm{EB}}\right) = \begin{pmatrix} V\mathbb{1} & \multicolumn{2}{c}{c\cos\theta^{(\mathrm{DH})}\sigma_z} \\ c\cos\theta^{(\mathrm{DH})}\sigma_z & \cos^2\theta^{(\mathrm{DH})}\left(V_{\mathrm{B}}+\delta_1-\frac{e^{2r}}{\cos^2\theta^{(\mathrm{DH})}}\right) & 0 \\ & 0 & \cos^2\theta^{(\mathrm{DH})}\left(V_{\mathrm{B}}+\delta_2-\frac{e^{-2r}}{\cos^2\theta^{(\mathrm{DH})}}\right) \end{pmatrix}. \tag{43}$$

Conditional entropy depends on squeezing parameter r, explicitly and through $\cos\theta^{(\mathrm{DH})}(r)$. For double homodyne detection, ideal measurements and three distinct cases of relationship between δ_i are possible, and their effect on joint entropy is illustrated by Fig. 4.

In the symmetrical case, $\delta_1 = \delta_2 \equiv \delta$ ($\sigma_1 = \sigma_2 \equiv \sigma$), $r_2 = -r_1$, so that choosing $r = 0$ is possible. Doing so results in vacuum ancilla, $x = 1$,and transmission becomes

$$\cos^2\theta \mapsto \frac{1}{\sigma}, \tag{44}$$

reproducing effect of $\Phi^{(\mathrm{H})}$, i.e. reducing channel transmission and channel noise by a factor of $\frac{1}{\sigma}$:

$$T \equiv T_{\mathrm{ch}} \cdot \frac{1}{\sigma}, \tag{45}$$

$$\xi \equiv \xi_{\mathrm{ch}} \cdot \frac{1}{\sigma}, \tag{46}$$

explicitly:

$$\frac{c}{\sqrt{\sigma}} = \sqrt{\frac{T_{\mathrm{ch}}(V^2-1)}{\sigma}} = \sqrt{T(V^2-1)} \equiv \tilde{c}, \tag{47}$$

$$\frac{V_{\mathrm{B}}+\sigma-1}{\sigma} = \frac{T_{\mathrm{ch}}(V-1)+\sigma+\xi_{\mathrm{ch}}}{\sigma} = T(V-1)+1+\xi \equiv \tilde{V}_{\mathrm{B}}, \tag{48}$$

$$\Phi^{\mathrm{DH}}\left(\Sigma_{\mathrm{AB}}^{\mathrm{EB}}\right) = \begin{pmatrix} V\mathbb{1} & \tilde{c}\sigma_z \\ \tilde{c}\sigma_z & \tilde{V}_{\mathrm{B}}\mathbb{1} \end{pmatrix}. \tag{49}$$

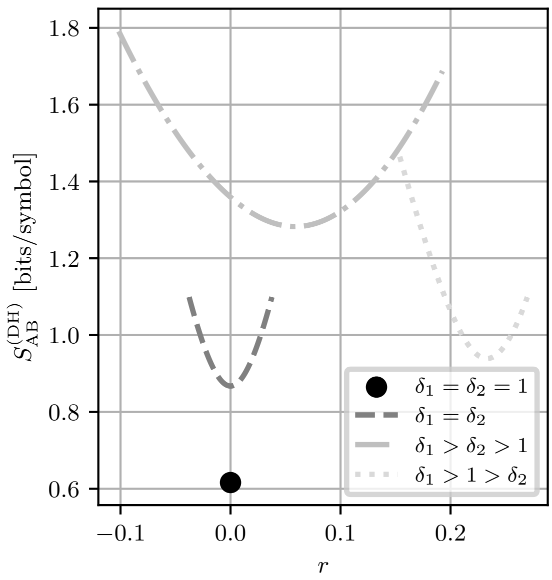


Figure 4: Joint entropy, $S_{\mathrm{AB}}^{(\mathrm{DH})}$, as a function of the squeezing parameter r at different values of δ_i. The channel parameters are: $V = 5$; $T_{\mathrm{ch}} = 0.95$; and $\xi_{\mathrm{ch}} = 10^{-2}$.

In this case, $r = 0$ is suboptimal for Eve, i.e. it minimizes Holevo information. For $r \neq 0$, $\cos^2\theta^{(\mathrm{DH})}$ becomes greater, increasing Holevo information.

In the asymmetrical case where $\delta_1 > 1$ and $\delta_2 > 1$, while setting $r = 0$ is possible, it results in greater Holevo information than in $r_{\min}$ that minimizes Holevo information.

Finally, if either $\delta_1 < 1$ or $\delta_2 < 1$, setting $r = 0$ is impossible. This case corresponds to high asymmetry of measurements.

Conditional entropy is calculated from symplectic eigenvalues of conditional covariance matrix $\Sigma_{\mathrm{A|B}}^{(\gamma)}$, which has the form:

$$\Sigma_{\mathrm{A|B}}^{(\gamma)} = V\mathbb{1} - c^2\sigma_z(V_{\mathrm{B}}\mathbb{1} + \Sigma_m^{(\gamma)})^{-1}\sigma_z. \tag{50}$$

For homodyne detection, we have

$$\nu_3^{(\mathrm{H})} = \sqrt{V\left(V - \frac{c^2}{V_{\mathrm{B}} + \sigma_N}\right)}, \tag{51}$$

for double homodyne detection,

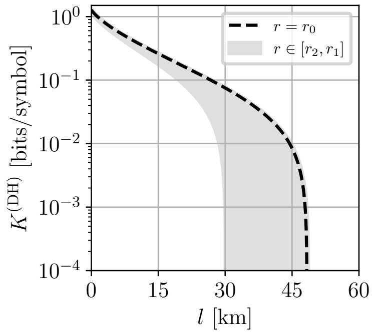
$$\nu_3^{(\mathrm{DH})} = V\sqrt{\frac{\left(V_{\mathrm{B}} + \delta_1 - \frac{c^2}{V}\right)\left(V_{\mathrm{B}} + \delta_2 - \frac{c^2}{V}\right)}{\left(V_{\mathrm{B}} + \delta_1\right)\left(V_{\mathrm{B}} + \delta_2\right)}}. \tag{52}$$

It is not possible to compute $r_{\max}$ that maximizes $\chi_{\mathrm{EB}}^{(\mathrm{DH})}$, see Eq. (26). However, due to equality between joint entropies at points r_2 and r_1 (see Fig. 8 and Fig. 10), it could be estimated numerically by calculating $\chi_{\mathrm{EB}}^{(\mathrm{DH})}(r_2), \chi_{\mathrm{EB}}^{(\mathrm{DH})}(r_1)$ in classical mixing channel and comparing them.

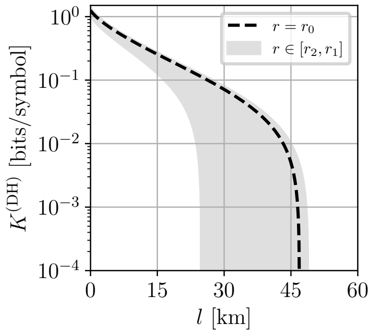
Comparing both detection schemes, see Fig. 7, difference in maximal transmission length in very slight, however bitrate in homodyne is lower.

3 Comparison between attenuator channel and classical mixing channel

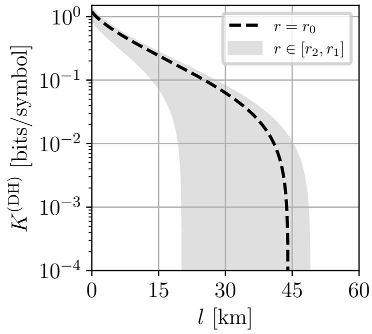
In Fig. 9a intervals do not intersect, confirmed by Fig. 10a. In Fig. 9b and 9c, intervals for $K^{(\mathrm{DH})}$ fully intersect due to relatively small δ, see Fig. 8b and 8c. Dependence on r, even for



(a) $q = 1$



(b) $q = 1.5$



(c) $q = 2$

Figure 5: Asymptotic secret fraction $K^{(\mathrm{DH})}$ versus channel length l, computed for fixed $\sigma_i^{(x)} = 1.05$ and various imbalance ratios q. Black dashed lines show results with fixed squeezing parameter $r = \frac{r_2+r_1}{2} \equiv r_0$, while green shaded areas indicate computed bounds for $K^{(\mathrm{DH})}$. The losses are 20 dB per 100 km. The parameters are: $V = 5$; $\beta = 0.95$; $\xi_{\mathrm{ch}} = 10^{-2}$.

symmetrical case, is greater than 15 km. Intervals will not intersect only in symmetrical case, see Fig. 11 and 10.

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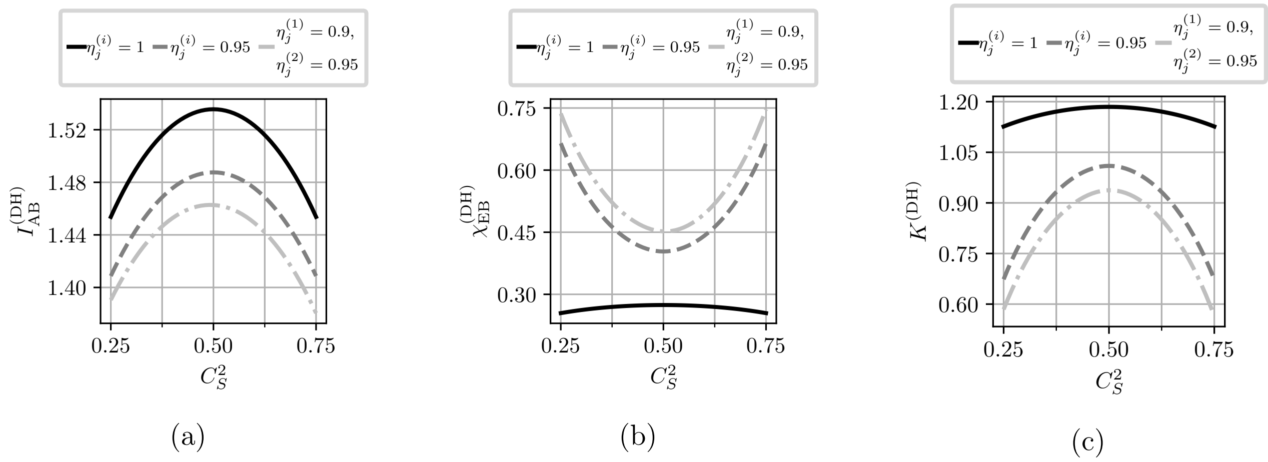


Figure 6: (a) Mutual information, $I_{\mathrm{AB}}^{(\mathrm{DH})}$, (b) Holevo information, $\chi_{\mathrm{EB}}^{(\mathrm{DH})}$ and (c) asymptotic secret fraction, $K^{(\mathrm{DH})}$, as a function of the signal beam splitter transmission, C_S^2, of the double homodyne receiver computed at different values of the detector efficiencies. All other beam splitters are assumed to be balanced and efficiencies not specified in the legend are taken to be unity. The parameters are: $V = 5$; $T_{\mathrm{ch}} = 0.95$; $\xi_{\mathrm{ch}} = 10^{-2}$; and $\beta = 0.95$.

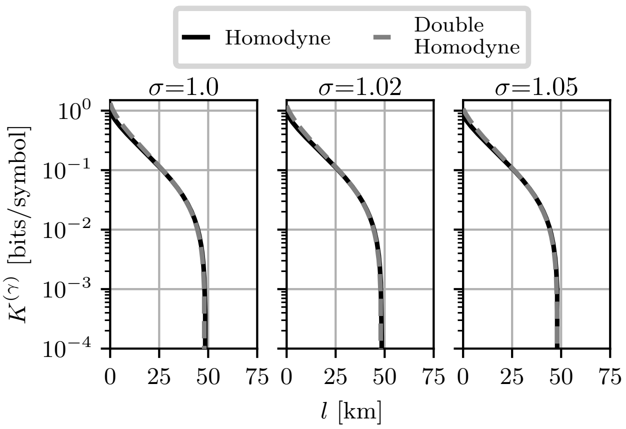
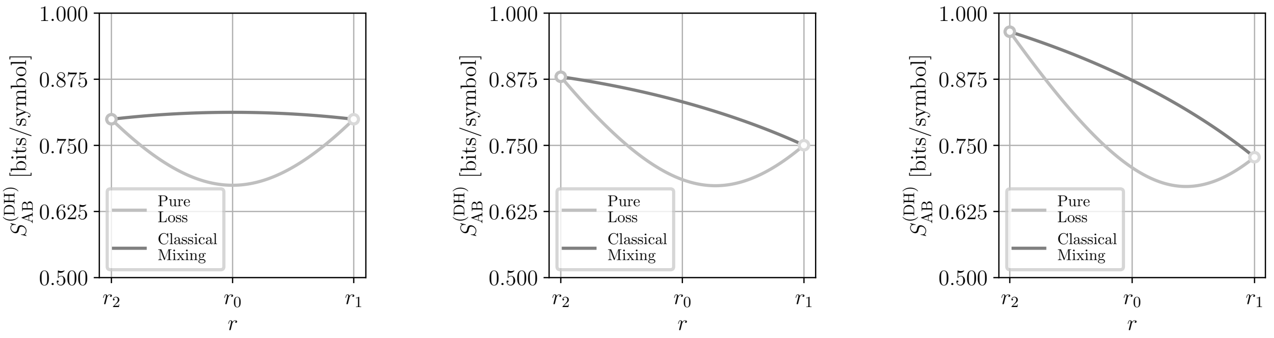
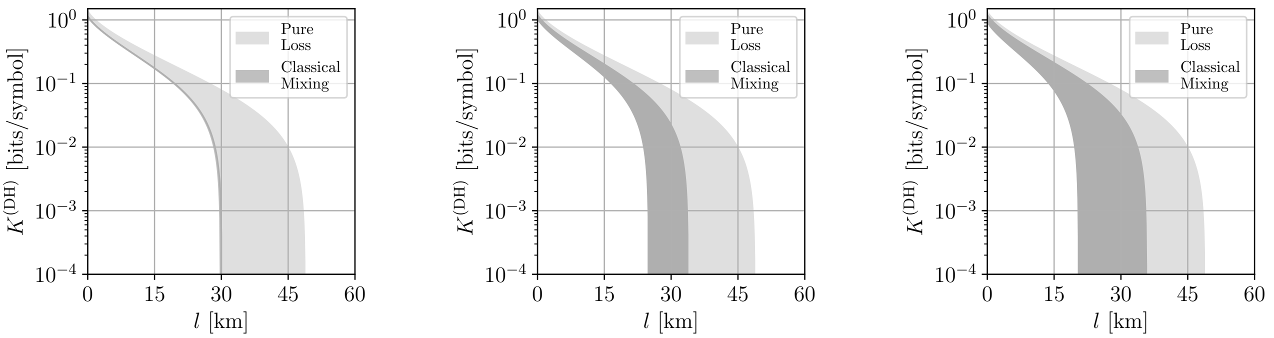


Figure 7: Asymptotic secret fraction $K^{(\gamma)}$ versus channel length l, computed for homodyne and double homodyne detection. Quadrature variances of both schemes are assumed to be equal, with value being indicated in panel title. The losses are assumed to be 20 dB per 100 km. The parameters are: $V = 5$; $\beta = 0.95$; $\xi_{\mathrm{ch}} = 10^{-2}$. For double homodyne, $r = 0$ (see main text).



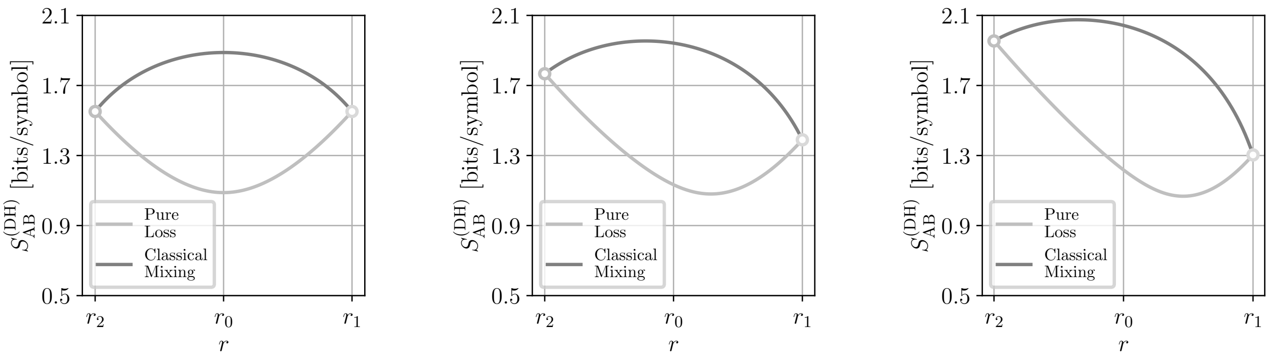
(a) $q = 1$ (b) $q = 1.5$ (c) $q = 2$

Figure 8: Joint entropy $S_{\mathrm{AB}}^{(\mathrm{DH})}$ as a function of r computed for the same parameters as in Fig. 11. $T_{\mathrm{ch}} = 0.95$.



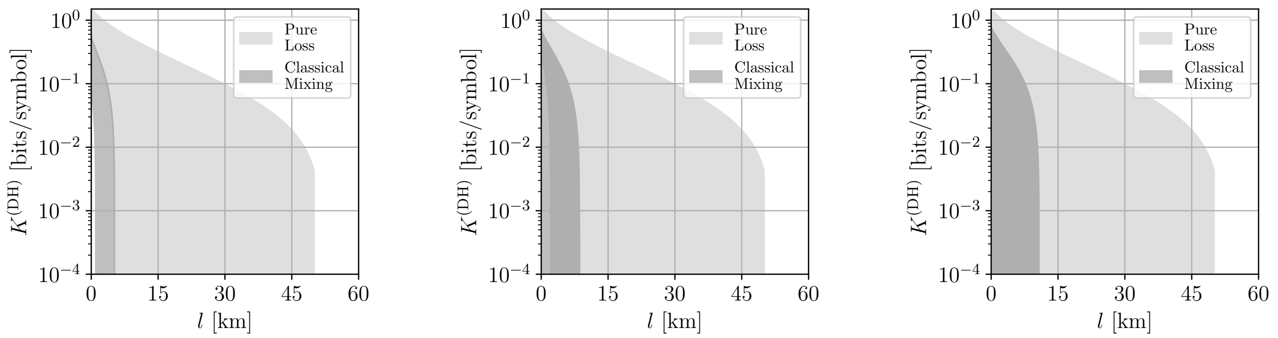
(a) $q = 1$ (b) $q = 1.5$ (c) $q = 2$

Figure 9: Asymptotic secret fraction $K^{(\mathrm{DH})}$ versus channel length l, computed for fixed $\sigma_i^{(x)} = 1.05$ and various imbalance ratios q. Green and blue shaded areas indicate computed bounds for $K^{(\mathrm{DH})}$, with greed for pure loss channel and blue for classical mixing channel. The losses are 20 dB per 100 km. The parameters are: $V = 5$; $\beta = 0.95$; $\xi_{\mathrm{ch}} = 10^{-2}$.



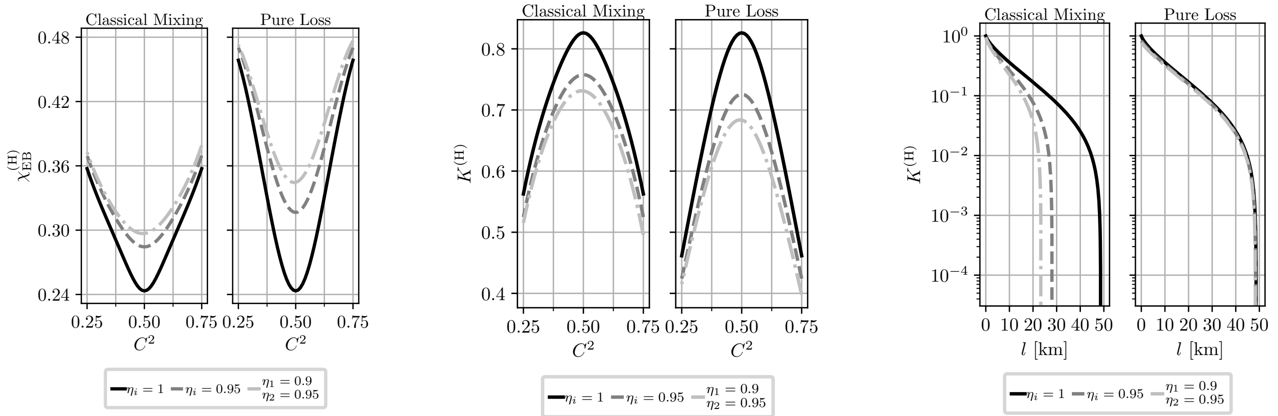
(a) $q = 1$ (b) $q = 1.5$ (c) $q = 2$

Figure 10: Joint entropy $S_{\mathrm{AB}}^{(\mathrm{DH})}$ as a function of r computed for the same parameters as in Fig. 11. $T_{\mathrm{ch}} = 0.95$.



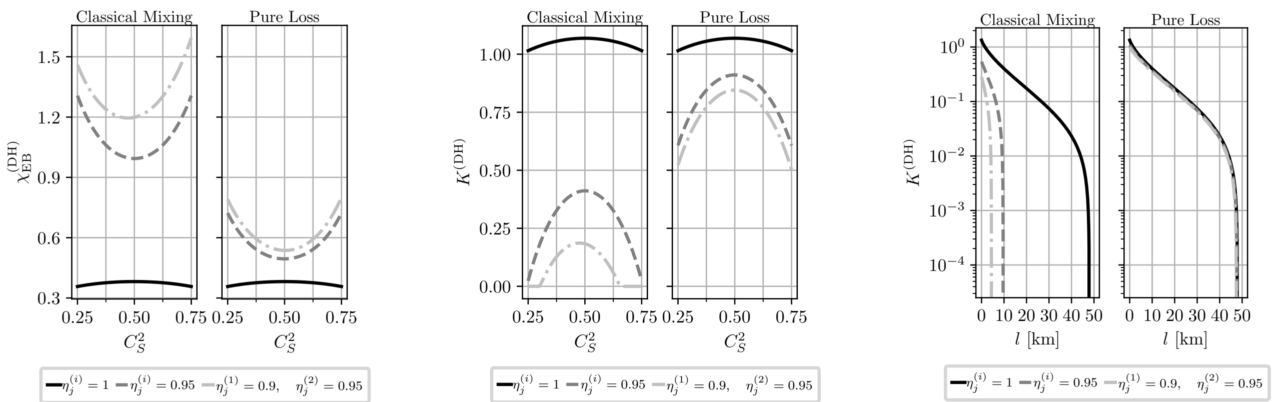
(a) $q = 1$ (b) $q = 1.5$ (c) $q = 2$

Figure 11: Asymptotic secret fraction $K^{(\mathrm{DH})}$ versus channel length l, computed for fixed $\sigma_i^{(x)} = 1.11$ and various imbalance ratios q. Green and blue shaded areas indicate computed bounds for $K^{(\mathrm{DH})}$, with greed for pure loss channel and blue for classical mixing channel. The losses are 20 dB per 100 km. The parameters are: $V = 5$; $\beta = 0.95$; $\xi_{\mathrm{ch}} = 10^{-2}$.



(a) (b) (c)

Figure 12: Comparison between pure loss and classical mixing channels. The losses are 20 dB per 100 km. The parameters are: $V = 5$; $\beta = 0.95$; $\xi_{\mathrm{ch}} = 10^{-2}$.



(a) (b) (c)

Figure 13: Comparison between pure loss and classical mixing channels. The losses are 20 dB per 100 km. The parameters are: $V = 5$; $\beta = 0.95$; $\xi_{\mathrm{ch}} = 10^{-2}$. For both cases $r = r_0$, which is suboptimal for pure loss and optimal for classical mixing.